

ENVS 193DS Final

Emilio Ojemann

2026-06-09

Table of contents

GitHub repository	2
Set Up	2
Problem 1. Research writing	2
a. Transparent statistical methods	2
b. Suggestions for rewriting	2
c. Further analysis	4
Problem 2. Data analysis	4
a. Clean and wrangle	4
b. Response variable	4
c. Purpose of study	4
d. Table of models	4
e. Running the models	4
f. Selecting the best model	4
g. Checking the diagnostics	4
h. Visualizing model predictions	4
i. Writing figure caption	4
j. Calculating model predictions	4
k. Interpreting results	4
Problem 3. Affective and exploratory visualizations	4
a. Comparing visualizations	4
b. Designing an analysis	4
c. Checking assumptions and running analysis	4
d. Creating a visualization	4
e. Writing a caption	4
f. Writing about results	4

g. Sharing affective visualization	4
--	---

GitHub repository

Set Up

Created repository, added description and README

```
# read in packages
library(tidyverse)
library(here)
library(janitor)

# read in nest observation data

# read in personal dinner cooking data
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a binomial generalized linear model (aka logistic). The response is a binary outcome: the probability of American Avocet microhabitat use (it either is used or it is not). The predictor is distance to water edge, so it is continuous. This makes sense to use a binomial GLM to test whether a continuous predictor variable relates to a binary response.

In part 2, they used a chi-square test. Both the response (habitat type, 5 levels) and the predictor (bird species, 3 levels) variables are categorical. It makes sense to use chi-square because they are seeing if a relationship exists between categories.

b. Suggestions for rewriting

Part 1: The probability that American Avocet's use microhabitat does not depend on how far the microhabitat is from the water's edge

Part 2: The

- c. Further analysis

Problem 2. Data analysis

- a. Clean and wrangle
- b. Response variable
- c. Purpose of study
- d. Table of models
- e. Running the models
- f. Selecting the best model
- g. Checking the diagnostics
- h. Visualizing model predictions
- i. Writing figure caption
- j. Calculating model predictions
- k. Interpreting results

Problem 3. Affective and exploratory visualizations

- a. Comparing visualizations
- b. Designing an analysis
- c. Checking assumptions and running analysis
- d. Creating a visualization
- e. Writing a caption
- f. Writing about results
- g. Sharing affective visualization